

Name:
Registration No:



MANIPAL UNIVERSITY
JAIPUR

Odd Semester Mid Term Examination, Academic Session: Jul-Nov 2026
Faculty of Science, Technology and Architecture, School of Computer Science and Engineering
Department of Artificial Intelligence and Machine Learning
B.Tech - CSE(AIML)

Course Code: AIM3141

Course: NoSQL Databases

Semester: V

Time: 1.5 hrs.

Max. Marks: 30

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is not allowed.

SECTION A

S.No.		Marks	CO
Q 1	Interpret the PACELC theorem by stating the trade-off made during normal operation when no network partition exists.	2	CO1
Q 2	Identify the purpose of the Computed Pattern in MongoDB and state one application scenario where it reduces repeated calculations.	2	CO2
Q 3	Outline the role of the \$facet stage in a MongoDB aggregation pipeline and state one practical analytical use.	2	CO3

SECTION B

Q 4	Differentiate horizontal partitioning, vertical partitioning, and consistent hashing. Associate one suitable distributed-workload condition with each strategy.	4	CO1
Q 5	Interpret the PACELC trade-off for two services: (i) a low-latency product-recommendation feed where slightly stale results are acceptable, and (ii) a payment-status service requiring the latest committed value. Justify the preferred latency-consistency choice for each.	(2+2)=4	CO1
Q 6	Apply the Outlier Pattern to an online-learning platform where most courses receive at most 50 reviews, while a few popular courses receive more than 100,000 reviews. Design a suitable MongoDB representation and justify the separation of outlier data.	4	CO2
Q 7	Construct a MongoDB aggregation pipeline that joins orders(customerId, total) with customers(_id, city) using \$lookup and \$unwind, and returns each order with the corresponding customer city.	3	CO3

SECTION-C

Q 8	Apply the Computed Pattern to a food-delivery dashboard that repeatedly displays each restaurant's total completed orders and average order value. Develop a representative MongoDB document fragment and describe the update strategy required to keep the computed fields consistent.	4	CO2
Q 9	Develop a MongoDB solution for a sensor_readings collection containing sensorId, timestamp, temperature, and humidity. (a) Construct a schema-validation rule that requires the fields and enforces suitable BSON data types. (b) Apply Atlas to connect to the database and retrieve sensor readings where temperature exceeds 35°C, displaying only sensorId, timestamp, and temperature.	(3+2)=5	CO3